Strict classical repeated-measures ANOVA for the biomarker-treatment

interaction: closed-form derivation and limitations

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1. Motivation

The founding pmsimstats question is the test of the biomarker-treatment interaction in an aggregated N-of-1 trial [@hendricksonOptimizing2020]. The pmsimstats-ng analysis layer fits a continuous-time mixed-effects model ($\text{nlme::lme}(\text{Sx} \sim \text{bm} + t + \text{Dbc} + \text{bm:Dbc}, \text{random} = \sim 1 \mid \text{ptID}, \text{correlation} = \text{corCAR1})$) and tests the interaction coefficient via a Wald t-statistic. A reviewer trained in classical analysis-of-variance methodology may reasonably ask whether a strict repeated-measures ANOVA (RM-ANOVA) test of the same interaction is available, what its closed-form expression looks like, and how it relates to the mixed-model test the framework currently produces. This white paper supplies that answer for the strict classical case.

The presentation is deliberately confined to **strict** RM-ANOVA: balanced split-plot design, categorical predictors, sphericity assumption, method-of-moments estimation. Modern extensions (the mixed-effects model for repeated measures, MMRM [@mallinckrodt2008recommendations]; generalised estimating equations, GEE [@liang1986longitudinal]; ANCOVA with random subject effects) are tabulated for context but are not strict RM-ANOVA in the textbook sense.

2. Notation and design

Consider the simplest design that supports a biomarker-treatment interaction test in a within-subject setting:

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomising the continuous biomarker at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomisation is the price of admission to the framework [@royston2006dichotomizing; @altman2006prognosis; @sennMastering2016].
- n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$ (active drug, placebo). In an N-of-1 design these are two phases through which every subject passes; in a parallel-group design they are between- subject and the design is not a split-plot.
- T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per subject per phase.

The biomarker stratum is **between-subjects** (each subject sits in exactly one stratum); the treatment phase and the timepoints are **within-subjects** (each subject sees both phases at every timepoint). This is the canonical *split-plot* repeated-measures design [@maxwellDelaney2017designing].

3. The strict RM-ANOVA model

Write Y_{ijlk} for the symptom outcome at biomarker stratum i, treatment phase j, subject l within stratum i, and timepoint k. The strict classical split-plot model is

$$\begin{aligned} Y_{ijlk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} \\ + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijlk}, \end{aligned} \tag{1}$$

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijlk} \sim N(0, \sigma_e^2)$. The model presupposes the classical sphericity (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2, every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

The biomarker-treatment interaction is the $(\alpha\beta)_{ij}$ term. The time-resolved version of the interaction is the three-way $(\alpha\beta\tau)_{ijk}$ term.

4. Sum-of-squares decomposition

The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \qquad (2)$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^{a} (\bar{Y}_{i\cdots} - \bar{Y}_{\cdots\cdot})^2, \qquad (3)$$

$$SS_{S(A)} = bT \sum_{i=1}^{a} \sum_{l=1}^{n} (\bar{Y}_{i\cdot l\cdot} - \bar{Y}_{i\cdots})^2. \qquad (4)$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the **biomarker $\times$ treatment interaction**, the target of the test), and a within-subjects error term $SS_{B\times S(A)}$:

$$SS_B = anT \sum_{j=1}^{b} (\bar{Y}_{\cdot j\cdot\cdot} - \bar{Y}_{\cdots\cdot})^2, \qquad (5)$$

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij\cdot\cdot} - \bar{Y}_{i\cdots} - \bar{Y}_{\cdot j\cdot\cdot} + \bar{Y}_{\cdots\cdot})^2, \qquad (6)$$

$$SS_{B\times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl\cdot} - \bar{Y}_{i\cdot l\cdot} - \bar{Y}_{ij\cdot\cdot} + \bar{Y}_{i\cdots})^2. \qquad (7)$$

The corresponding degrees of freedom are $df_A = a-1$, $df_{S(A)} = a(n-1)$, $df_B = b-1$, $df_{AB} = (a-1)(b-1)$, and $df_{B\times S(A)} = a(n-1)(b-1)$ (the latter being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor).

5. The test statistic: closed form

The biomarker-treatment interaction is tested by the F-ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B\times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B\times S(A)}/df_{B\times S(A)}}. \qquad (8)$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B\times S(A)})$.

5a. The 2×2 split-plot case

Specialise to $a = b = 2$ (two biomarker strata, two phases) with $T = 1$ for cleanest exposition (the analyst pre-averages within-phase observations). The sample interaction contrast is

$$\hat{L} \;=\; (\bar{Y}_{11\cdot} - \bar{Y}_{12\cdot}) - (\bar{Y}_{21\cdot} - \bar{Y}_{22\cdot}), \qquad (9)$$

i.e. the difference of within-subject treatment differences across biomarker strata. This is the natural biomarker-treatment interaction statistic.

The interaction parameter under the sum-to-zero ANOVA parameterisation satisfies $(\widehat{\alpha\beta})_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares is

$$SS_{AB} \;=\; nT \cdot 4 \cdot (\hat{L}/4)^2 \;=\; \frac{nT\,\hat{L}^2}{4}. \qquad (10)$$

With $df_{AB} = 1$, the interaction mean square equals the interaction sum of squares.

The within-subjects error mean square is the pooled estimator of σ_e^2 from the treatment-by-subject-within-stratum component of variance. Under the sphericity assumption and balanced data, $MS_{B\times S(A)} = \hat{\sigma}_e^2$.

Substituting into equation (8):

$$\boxed{F_{AB} \;=\; \frac{nT\,\hat{L}^2}{4\,\hat{\sigma}_e^2}}, \qquad df_1 = 1, \quad df_2 = 2(n-1). \qquad (11)$$

This is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.

5b. Within-subject contrast representation

Equation (11) is equivalent to a two-sample t-test on the within-subject difference. Define $\Delta_{il} = \bar{Y}_{i1l\cdot} - \bar{Y}_{i2l\cdot}$ as the within-subject treatment difference for subject l in biomarker stratum i, averaged over T within-phase timepoints. Then

$$\hat{L} \;=\; \bar{\Delta}_1 - \bar{\Delta}_2, \qquad \bar{\Delta}_i \;=\; \frac{1}{n}\sum_{l=1}^{n} \Delta_{il}. \qquad (12)$$

Under sphericity, Δ_{il} are iid within stratum with $\mathrm{Var}(\Delta_{il}) = 2\sigma_e^2/T$ (the random subject effect cancels in the within-subject difference). The variance of $\hat{L}$ is therefore

$$\mathrm{Var}(\hat{L}) \;=\; \frac{2 \cdot 2\sigma_e^2}{nT} \;=\; \frac{4\sigma_e^2}{nT}. \qquad (13)$$

The Welch-style t-statistic for the contrast is

$$t_{AB} \;=\; \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} \;=\; \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \qquad (14)$$

and $t_{AB}^2 = F_{AB}$, recovering equation (11). The strict classical RM-ANOVA F-test for the biomarker-treatment interaction is therefore identical to a two-sample t-test on the within-subject treatment differences, with degrees of freedom $2(n-1)$. This identity is well known in the split-plot literature [@maxwellDelaney2017designing, ch.\ 12; @kirk2013experimental] and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

5c. Generalisation: a strata, b phases

For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (8) generalises to

$$F_{AB} = \frac{SS_{AB}/[(a-1)(b-1)]}{SS_{B\times S(A)}/[a(n-1)(b-1)]}. \qquad (15)$$

Under sphericity, $F_{AB} \sim F((a-1)(b-1), a(n-1)(b-1))$ under H_0. The closed form of the interaction sum of squares follows directly from equation (6).

5d. The time-resolved three-way case

Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$. The F-statistic is

$$F_{AB\tau} = \frac{MS_{AB\tau}}{MS_{B\tau\times S(A)}}, \qquad (16)$$

with $df_1 = (a-1)(b-1)(T-1)$ and $df_2 = a(n-1)(b-1)(T-1)$. The sums of squares follow the standard split-plot pattern [@maxwellDelaney2017designing, ch.\ 14].

6. Sphericity, Mauchly, and the Greenhouse-Geisser correction

Equations (11) and (15) presuppose sphericity: the within-subject covariance matrix is compound-symmetric. When this assumption fails, the nominal F-distribution is no longer exact under H_0, and the test inflates Type I error. Two diagnostics and two corrections are standard:

- **Mauchly's test** [@mauchly1940significance]. Likelihood-ratio test for compound symmetry against an unstructured alternative. Often rejects in moderate samples; not informative about the direction of departure.
- **Box's** ϵ [@box1954someproblemsI; @box1954someproblemsII]. A scalar measure of departure from sphericity, with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T-1)$ at maximum departure.
- **Greenhouse-Geisser correction** [@greenhouseGeisser1959]. Estimate $\hat{\epsilon}$ from the observed within-subject covariance and adjust degrees of freedom: $F_{AB} \sim F(\hat{\epsilon}\cdot df_1, \hat{\epsilon}\cdot df_2)$ under H_0. Conservative.
- **Huynh-Feldt correction** [@huynhFeldt1976]. Less conservative variant, recommended when $\hat{\epsilon} > 0.75$.

The pmsimstats DGP imposes AR(1) within-factor correlation, which is **not** compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and the Greenhouse-Geisser correction substantially attenuates the available degrees of freedom in F-tests on within-subject effects. In simulation, the strict- RM-ANOVA Type I error in the present setting is documented at 13–17 percent for the CO and OL designs absent correction (audit at `docs/06-ar1-residual-correlation.tex`); after Greenhouse-Geisser correction the Type I error returns toward nominal but at the cost of substantial power. The mixed-model analysis with `corCAR1` does not pay this cost because it models the AR(1) structure directly.

7. Power calculation under the closed form

For the 2×2 case, equation (11) yields a clean closed-form power formula. Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT\,\hat{L}^2/(4\hat{\sigma}_e^2)$ follows a non-central F distribution with non-centrality parameter

$$\lambda \;=\; \frac{nT\,\delta^2}{4\sigma_e^2}. \qquad (17)$$

Power at level α is

$$1-\beta \;=\; \Pr\Big[F_{AB} > F_{1,2(n-1),1-\alpha} \;\Big|\; \lambda\Big], \qquad (18)$$

which is computable via standard non-central F functions (`stats::pf` in R with the `ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (17) inverts to

$$n \;=\; \frac{4\sigma_e^2\lambda}{T\delta^2}, \qquad (19)$$

with λ chosen to deliver the target power at the design α.

This formula assumes (i) sphericity, (ii) categorical biomarker at the dichotomised cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, (v) iid within- phase observations after within-subject differencing. All five assumptions are violated to varying degrees in the pmsimstats DGP; see §8.

8. Limitations of the strict classical formulation

The closed-form simplicity of equation (11) is bought with five strong assumptions, each of which is empirically problematic in the pmsimstats setting.

1. **Categorical biomarker.** The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomising at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error [@royston2006dichotomizing; @altman2006prognosis; @sennMastering2016]. The information loss can be substantial: in the carryover-sensitivity production data the median split gives a $\widehat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous-biomarker mixed-model coefficient, translating to a 30 to 50 percent power penalty.

2. **Sphericity.** Compound-symmetric within-subject covariance is empirically rejected for time-series symptom data: nearby timepoints are more correlated than distant ones. The AR(1) structure that pmsimstats implements explicitly violates sphericity. After Greenhouse-Geisser correction, the test recovers nominal Type I error but loses degrees of freedom and power.

3. **No carryover model.** Strict RM-ANOVA has no provision for residual drug effect during washout; the model treats off-drug observations as if no drug had been administered. Under the pmsimstats DGP with $t_{1/2} = 1.0$ week, this misspecification biases $\widehat{L}$ toward zero (attenuation by 30 to 35 percent in the production data, comparable to the A1 specification evaluated in the carryover-sensitivity manuscript).

4. **Balanced design required.** Equations (11) and (15) assume equal stratum sizes and equal numbers of within-phase timepoints. Real trials have missing data; classical RM-ANOVA handles missingness only through complete-case deletion. The MMRM framework's likelihood-based handling of MAR data [@mallinckrodt2008recommendations] is genuinely better-behaved.

5. **Equal time spacing.** Strict RM-ANOVA treats time as a categorical factor with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework cannot do.

9. Relationship to the pmsimstats MMRM analysis

The strict-RM-ANOVA F-statistic in equation (11) and the pmsimstats `nlme::lme` Wald t-statistic on `bm:Dbc` are **asymptotically equivalent** under three conditions: (i) the biomarker is dichotomised at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound- symmetric, and (iii) `Dbc` is binary (the A1 specification rather than the matched-decay A2 specification). When all three hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modelled directly, and the continuous `Dbc` exposure-decay predictor (A2) leverages the off-drug trajectory.

For the founding pmsimstats interaction question [@hendricksonOptimizing2020], the strict-RM-ANOVA test is therefore a **conservative reduction** of the available analysis rather than a competing alternative. Where a reviewer requests "RM-ANOVA results", the natural response is one of (a) point at the existing MMRM analysis as a continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from equation (11) as a sensitivity comparator under dichotomised biomarker and binary phase, or (c) report both with the explicit power-loss tabulation.

10. Recommendation for pmsimstats

We recommend retaining the MMRM-with-continuous-time analysis as primary and adding a strict-RM-ANOVA comparator as a planned sensitivity:

1. **Categorise the biomarker at the median.** Conventional choice; the alternative pre-specified clinical cut-point is acceptable when one exists.
2. **Pre-average within-phase timepoints to a single mean per subject per phase.** Reduces the T-axis to $T = 1$ for the sensitivity test.
3. **Compute $\hat{L}$ per equation (9) and F_{AB} per equation (11).** This is computable by hand from the four cell means and the within-subject error mean square.
4. **Report the strict-RM-ANOVA p-value alongside the MMRM p-value in the manuscript table.** The two should be in the same order of magnitude. Substantial divergence between them indicates that the dichotomisation, the sphericity assumption, or the binary phase indicator is doing meaningful work, and the discrepancy is informative for interpretation.

The carryover-sensitivity manuscript at `analysis/report/carryover-sensitivity/` already evaluates the A1 binary-phase specification, which is the analysis-model counterpart to the strict-RM-ANOVA closed form derived here. The manuscript reports the A1 specification's bias and coverage under carryover. The strict-RM-ANOVA test inherits the A1 specification's attenuation properties.

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